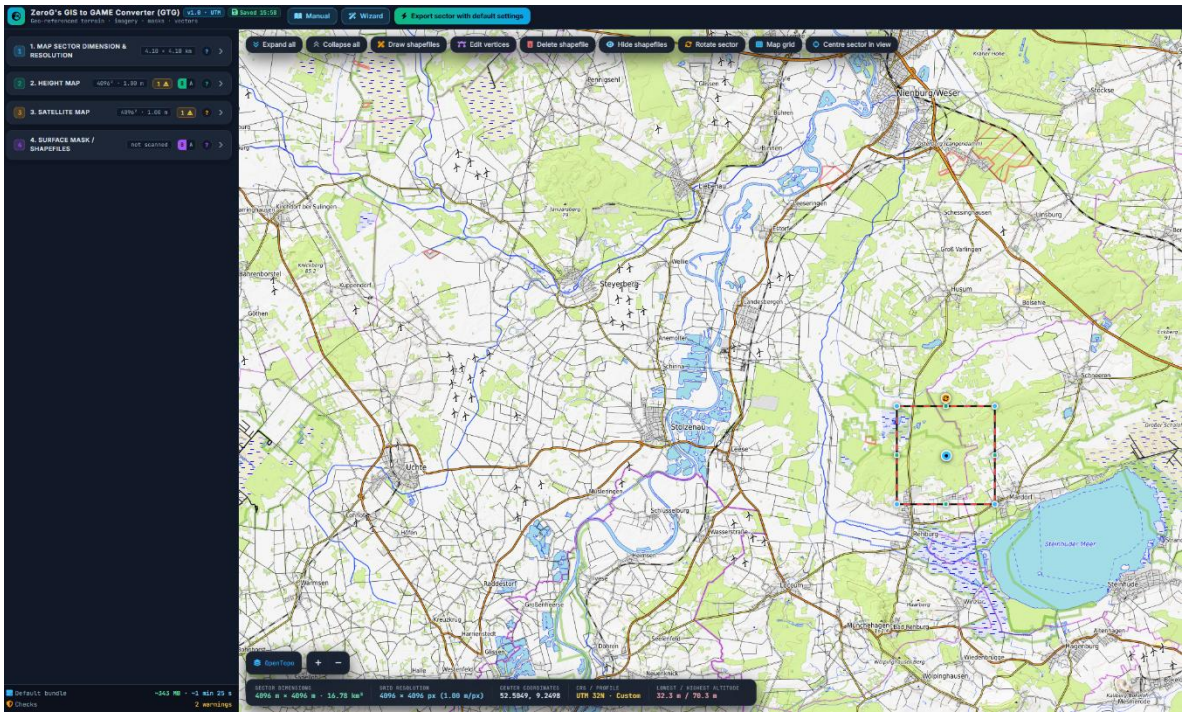
ZeroG's GIS TO GAME Converter (GTG)

User manual · version 1.0 · November 2026



Real-world terrain for game engines – heightmaps, satellite maps, surface masks, shapefiles and terrain shaping, georeferenced in UTM and ready for Arma Reforger, Arma 3, Unreal Engine and Unity.

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Preface

ZeroG's GIS to Game Converter (GTG) turns any place on Earth into game-ready terrain data. You pick a sector on a map, and the tool produces a georeferenced heightmap, a satellite image, surface masks for material painting, shapefiles of roads, rivers, forests and buildings, and a reshaped ("morphed") heightmap in which roads are graded, rivers are carved and buildings stand on smoothed ground. Optional special zones add craters, trenches, berms, dams, pits, dunes and other features on top.

The tool targets map makers for Arma Reforger (Enfusion), Arma 3 (Terrain Builder), Unreal Engine and Unity, and anyone who needs real-world terrain in a metric grid. Everything runs inside your web browser: there is nothing to install, no account and no server of its own. Data is loaded directly from public services (OpenStreetMap, AWS Terrain Tiles, Sentinel-2 cloudless, OpenTopoMap) or from your own files and WMS servers.

Core facts at a glance

Topic	Fact
Delivery	One HTML file, opens in Chrome or Edge. Internet access is needed for map data.
Coordinate system	WGS84 / UTM (zone of the sector centre). The south-west corner of the sector is game origin 0,0.
Sector size	128 m up to 65,536 m per side; square or rectangular; optional rotation against the UTM grid.
Heightmap	Cell size 0.25–8 m, max 8,192 px per side. Formats .asc, 16-bit .png, 16-bit .raw, 32-bit .r32, float GeoTIFF.
Elevation sources	AWS Terrain Tiles (global), custom elevation WMS, local DEM file (.tif / .asc).
Imagery sources	Sentinel-2 cloudless (2016 = commercial use allowed), OpenTopoMap, custom XYZ tiles, custom WMS.
Vectors	OpenStreetMap scan in 8 categories / 33 subtypes, plus your own drawn or imported shapes. Shapefile and GeoJSON.
Terrain shaping	Water carving, road and rail grading, building smoothing, flatten areas, 19 special-zone types.
Surface masks	Colour-coded mask with legend, grayscale splat masks per category or subtype, RGBA splatmap, Enfusion block check.
Engine profiles	Unreal Engine 5, Unity, Arma 3, Arma Reforger – one click sets grid, formats and resolutions.
Languages	English (default), German, French, Spanish, Italian, Polish, Danish, Swedish, Norwegian, Czech, Ukrainian, Russian.
Storage	Session, bookmarks, history (20 versions) and tile cache are kept in the browser. Projects can be saved as .json.

How this manual is organised

- **Part 1 – Getting started:** the screen layout, the header, the map toolbar and the concepts that apply everywhere (Basic / Advanced, messages, help).
- **Part 2 – Tutorial:** a complete project from an empty map to the exported bundle, step by step with pictures.
- **Part 3 – Feature reference:** every section and every control, grouped exactly like the side panel.
- **Part 4 – Reference:** keyboard shortcuts, engine import notes, limits, troubleshooting, credits and licences.

> **Note Screenshots** in this manual were taken with a sample terrain near Lemnos (Greece). Colours of map tiles may differ from what you see with live data; the controls are identical.

Part 1 – Getting started

1.1 Starting the tool

- Open the HTML file in a current Chrome or Edge browser (double-click or drag it into a browser window).
- The tool starts in English with all sections folded and a 4,096 × 4,096 m sector at 1 m cell size.
- Your last session is restored automatically on the next start (“Restored” in the header).
- A private window starts clean – useful for a fresh project or for testing.

1.2 The screen at a glance

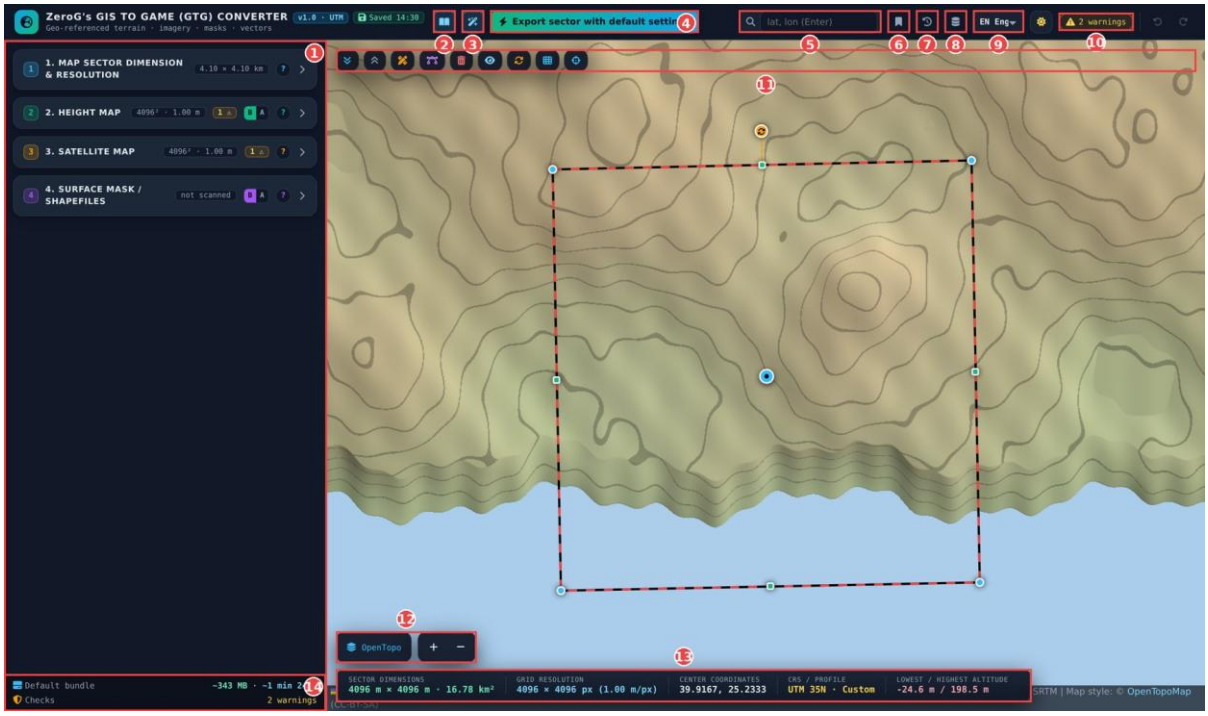


Figure 1 – *Main screen with the numbered areas explained below*

No.	Area	Purpose
1	Side panel	Sections 1–4 with all settings, previews and exports. Click a header to fold or unfold it.
2	Manual	Opens the built-in manual (can be printed or saved as PDF).
3	Wizard	Step-by-step tour that highlights every visible control.
4	Default export	One-click ZIP with heightmap, satellite map, masks, shapefiles and morphed heightmap.
5	Search	Enter coordinates “lat, lon” and press Enter – the sector jumps there.
6	Locations	Bookmarks and recently used sectors.
7	History	Saved project versions; one is stored with every export.
8	Data cache	Tile and OpenStreetMap cache, size limit and clearing.
9	Language	Interface language (12 languages).

No.	Area	Purpose
10	Messages	Number of errors and warnings; click to see all and jump to the setting.
11	Map toolbar	Drawing, editing, deleting, show / hide, rotate, map grid, centre sector.
12	Map controls	Background map, custom WMS and zoom.
13	Status bar	Sector size, grid, centre, CRS / profile / rotation, lowest and highest altitude.
14	Footer	Expected size and time of the default bundle and the overall check status.

1.3 Header bar



Figure 2 – *Header bar*

- **Version and save tag –** shows “Saved hh:mm” after each automatic save.
- **Undo / Redo –** right end; reverses shape edits, deletions and on/off changes (Ctrl+Z / Ctrl+Y).
- **Bright / dark –** sun icon switches the colour theme; the choice is saved.

1.4 Map toolbar

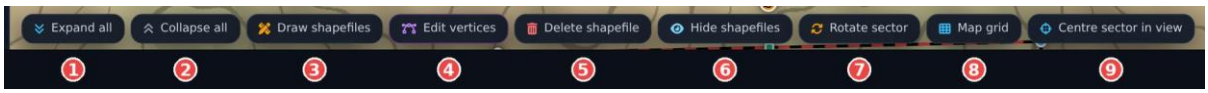


Figure 3 – *Map toolbar (labels are shown as tooltips on smaller screens)*

No.	Button	What it does
1	Expand all	Opens every section.
2	Collapse all	Folds every section.
3	Draw shapefiles	Panel with class list and Polygon / Line / Rectangle tools.
4	Edit vertices	Move, add (+) or delete (Del) vertices of any shape. Key E.
5	Delete shapefile	Delete mode: every clicked shape is removed (Undo works).
6	Show / hide shapefiles	Hides all shapes on the map; they are still exported.
7	Rotate sector	Angle input, slider, “Align to map grid” / “Align to UTM grid”.
8	Map grid	UTM, lat / lon or sector grid overlay (display only).
9	Centre sector in view	Moves the sector to the map centre; size and settings stay.

1.5 Concepts used everywhere

Basic and Advanced

- Sections 2 and 4 have a small B | A switch next to the “?” button.

- Basic shows the essential controls; hidden settings keep their values and are summarised in one line.
- Advanced shows every option. In section 4, subsections 4.2, 4.2b and 4.3 appear only in Advanced.
- Sections 1 and 3 always show all options.

Messages and checks

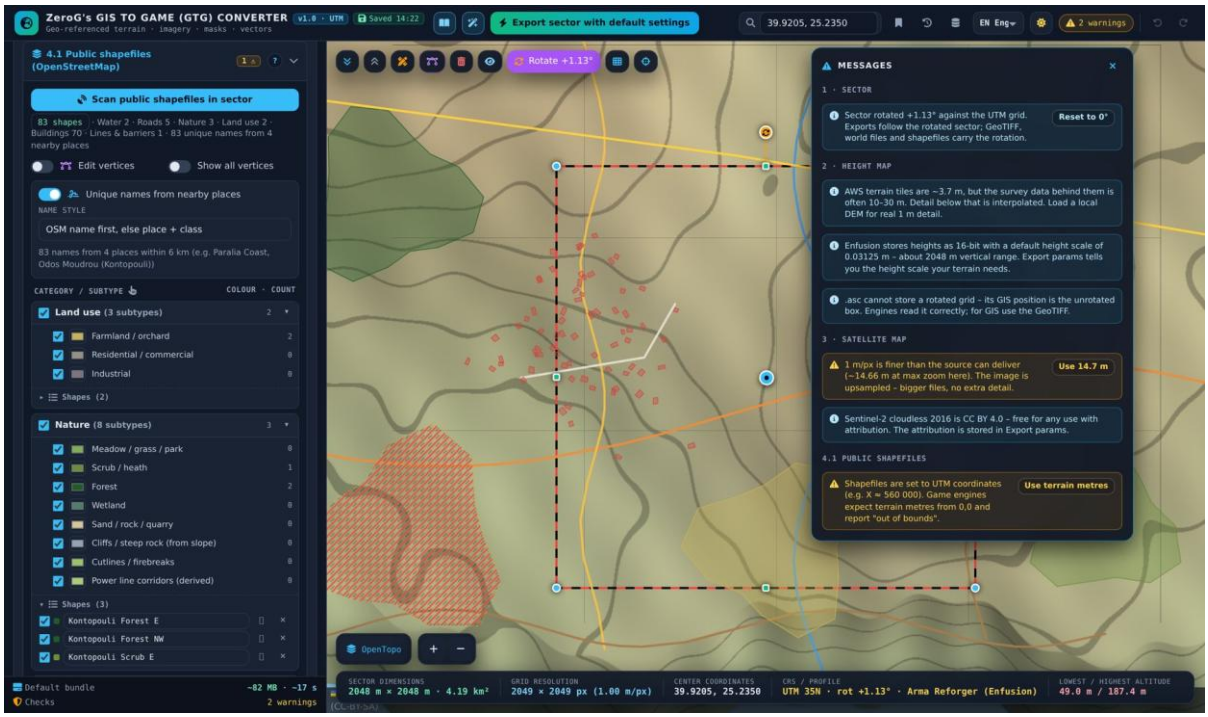


Figure 4 – *Message list – click a message to jump to the setting, use the button for a one-click fix*

- Every change is checked live. Red = error (blocks the export), amber = warning, blue = tip.
- The header button shows the count; the coloured badges on section headers open the same list.
- Clicking a message opens the section, scrolls to the setting and highlights it.
- Many messages offer a fix button (e.g. “Use terrain metres”, “Use 2016”, “Make square”).

Help, wizard and built-in manual

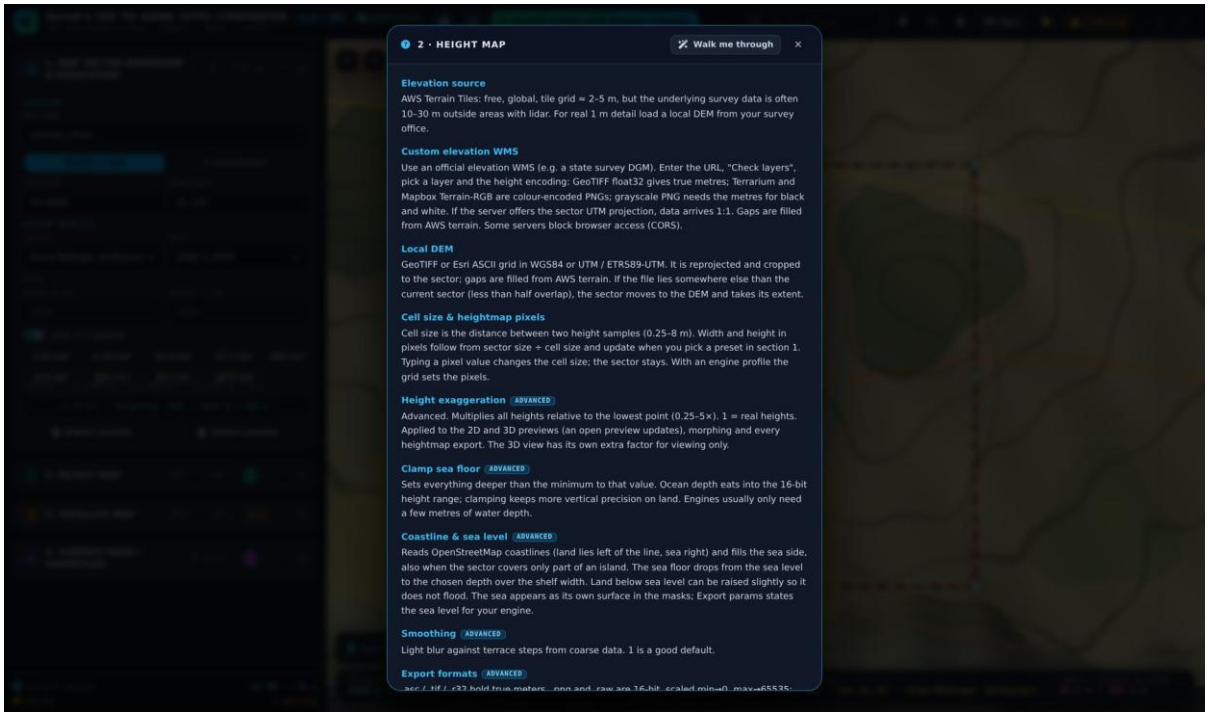


Figure 5 – *Section help ("?") with the "Walk me through" button*

- **"?" on a section –** explains every control of that section; "Walk me through" starts the wizard there.
- **Wizard –** black panel at the right, ← / → to move; jumps between sections with the drop-down.
- **Tooltips –** hover any control for a one-line explanation.



Figure 6 – *Wizard highlighting a control*

Part 2 – Tutorial: a WW1 front line on Lemnos for Arma Reforger

Goal: a 2 × 2 km Arma Reforger terrain of a coastal hillside, aligned to the map grid, with real roads, forests and buildings, a shelled no-man's-land, two trench lines, a bomb crater line, a machine-gun nest and a mine crater. At the end you export one ZIP with everything the Enfusion Workbench needs.

> **Note** Time needed: about 10 minutes. Result: heightmap, morphed heightmap, satellite map, surface masks per surface, shapefiles in terrain metres and the import values in params.json.

Step 1 – Find the place

- Type the coordinates "39.9205, 25.2350" into the search box and press Enter.
- The red-black dashed frame (the sector) jumps there. Rename the map to "Lemnos_Front" in section 1 – this becomes the prefix of all files.
- Alternatives: drag the blue centre marker, pan the map and click "Centre sector in view", or enter four coordinates.



Figure 7 – *Sector placed via coordinate search*

Step 2 – Choose the engine profile and size

- Open section 1. Set Target to “Arma Reforger (Enfusion)” and Grid to “2049 × 2049”.
- The profile sets the heightmap formats (.asc + 16-bit .png), per-subtype splat masks, the block check and terrain-metre coordinates.
- With a 1 m cell size the sector becomes 2,048 × 2,048 m (2049 vertices = 2048 faces = 64 blocks of 32 faces).



Figure 8 – *Arma Reforger profile with a 2049 grid*

Step 3 – Align the sector to the map grid

- Click “Rotate sector” on the map toolbar and then “Align to map grid”.
- The sector turns by the grid convergence (+1.13° here), so its edges run exactly north–south and east–west on the map.
- Switch on “Map grid → Sector” to see the game grid from the south-west corner (0,0). It is display only.

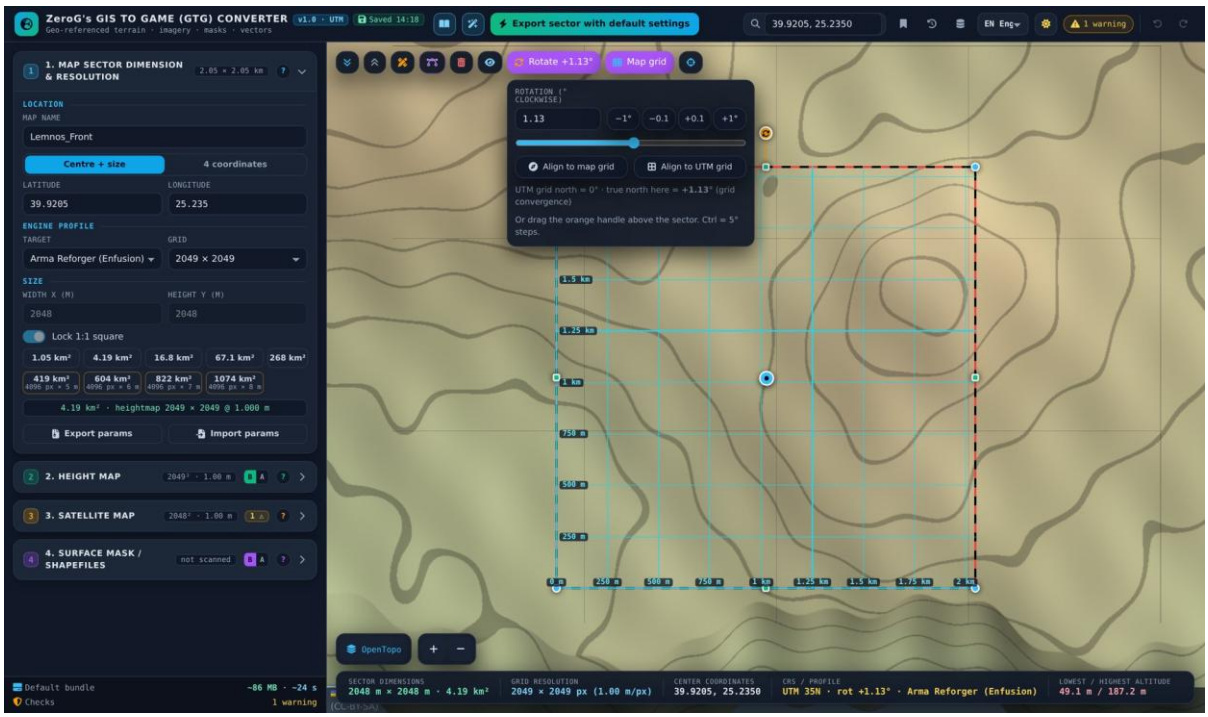


Figure 9 – *Rotation panel and sector grid*

Step 4 – Check the terrain

- Open section 2 and click Preview. The window compares the heightmap (right) with the satellite image (left) – drag the orange handle.
- Use the shader bar: Relief for an overview, Contours for heights, Slope for steepness.
- The status line shows the elevation range – important for the Enfusion height scale.

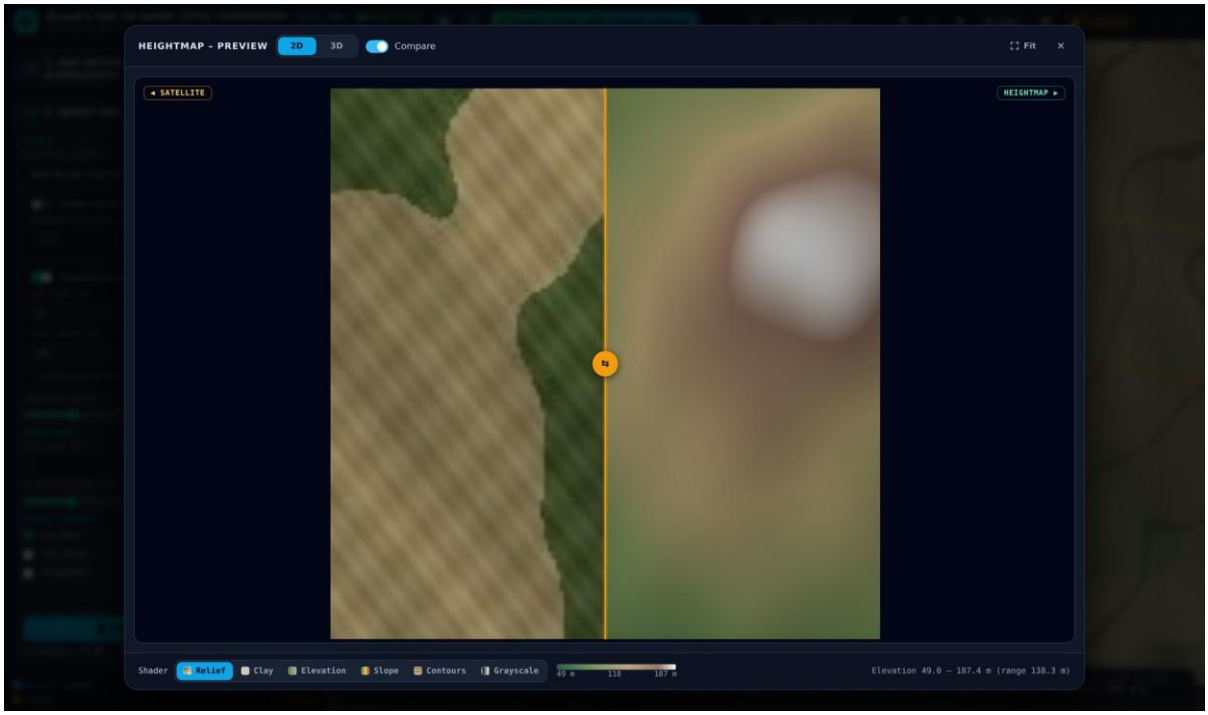


Figure 10 – *Heightmap preview with compare slider*

Step 5 – Check the satellite image

- Section 3: keep “Sentinel-2 cloudless 2016” (commercial use allowed with attribution).
- Preview shows the image against the terrain relief. A finer resolution than the source only makes files bigger.

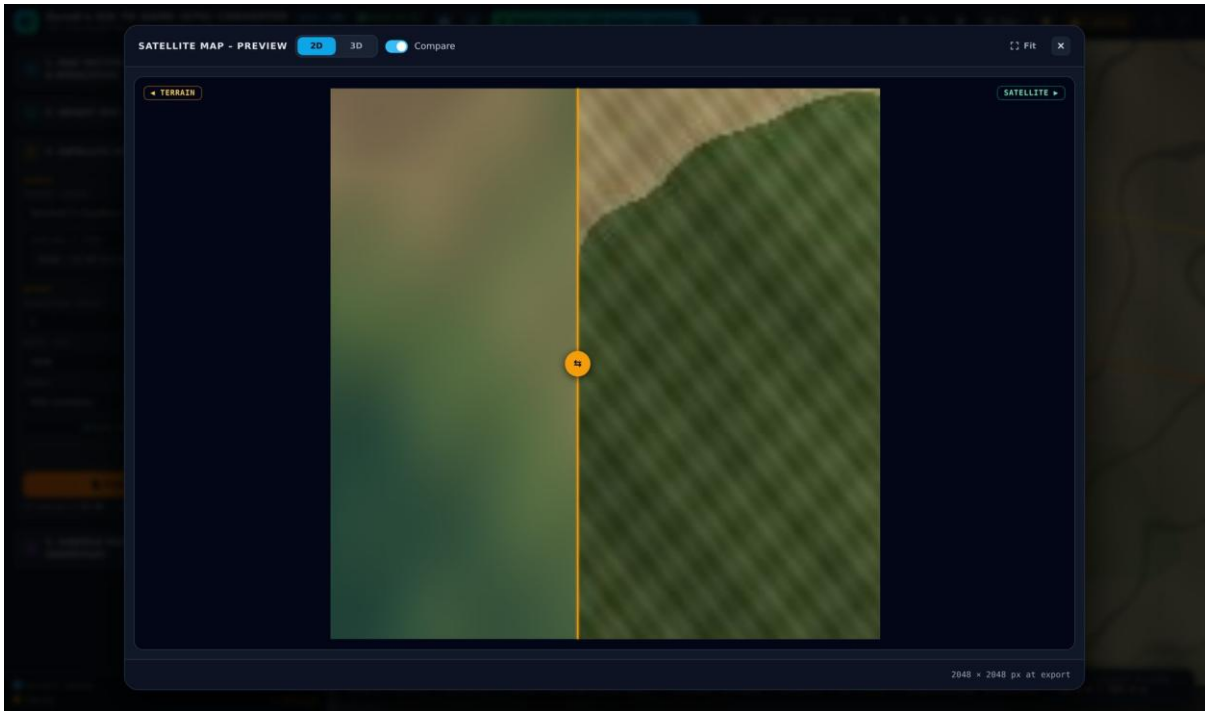


Figure 11 – *Satellite preview*

Step 6 – Load and clean the public shapefiles

- Section 4 → 4.1 → “Scan public shapefiles in sector”. Roads, rails, water, land use, forest, buildings and power lines are loaded from OpenStreetMap.
- Click a field you do not want: it gets red diagonal bars and is excluded from all exports.
- Open “Shapes (n)” under a category to rename shapes, exclude, zoom or delete them.



Figure 12 – *Scanned shapes; the excluded farmland shows red bars*

Step 7 – Declare the special zones

- Switch section 4 to Advanced (A) and open 4.2b. Draw a polygon zone over no-man's-land and set its type to "Artillery field".
- Draw two line zones as "Trench / earth crack", one as "Bomb crater line", a small polygon as "Firing position / revetment" and one as "Explosion".
- Each zone has its own card with intensity, radius, edge feather, "Keep roads & terrain edits" and a dice for a new random layout.



Figure 13 – *Special zones on the map and their cards in 4.2b*

Step 8 – Morph the terrain

- Open 4.3 and click Preview. Left = original, right = morphed: roads graded, buildings smoothed, craters and trenches cut.
- Switch to the Slope shader to judge steep edges, or to Clay for the pure shape.

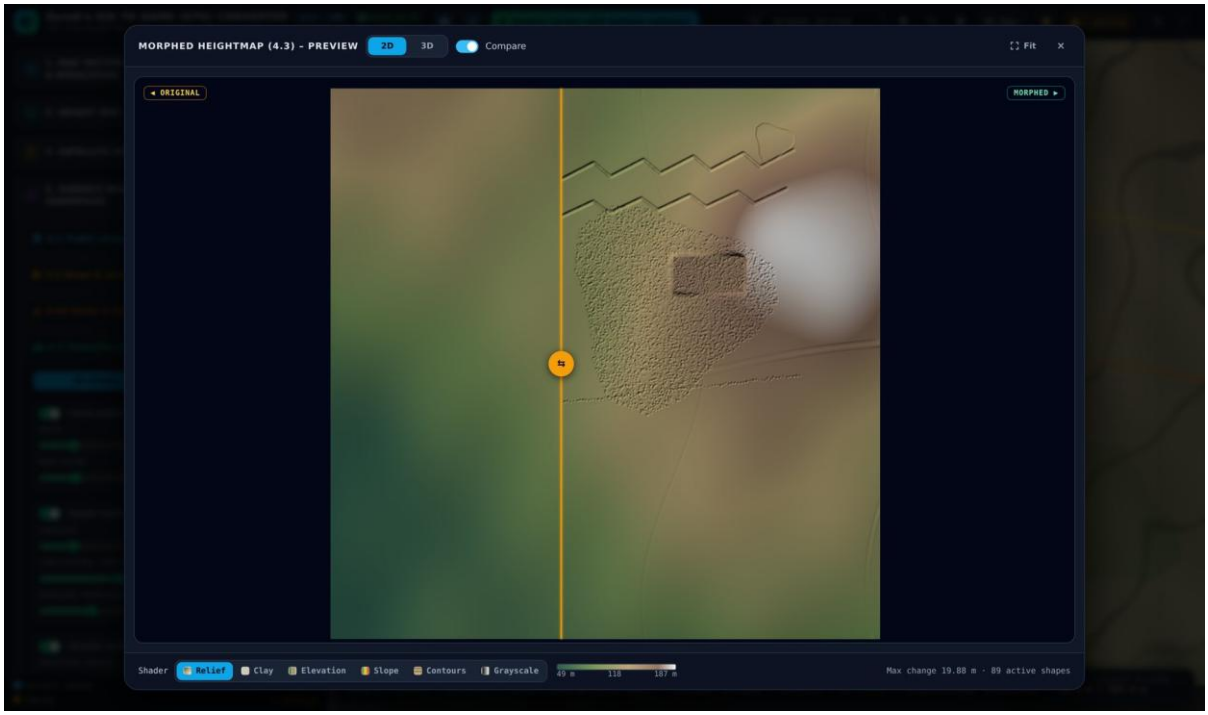


Figure 14 – *Before / after preview of the morphed heightmap*

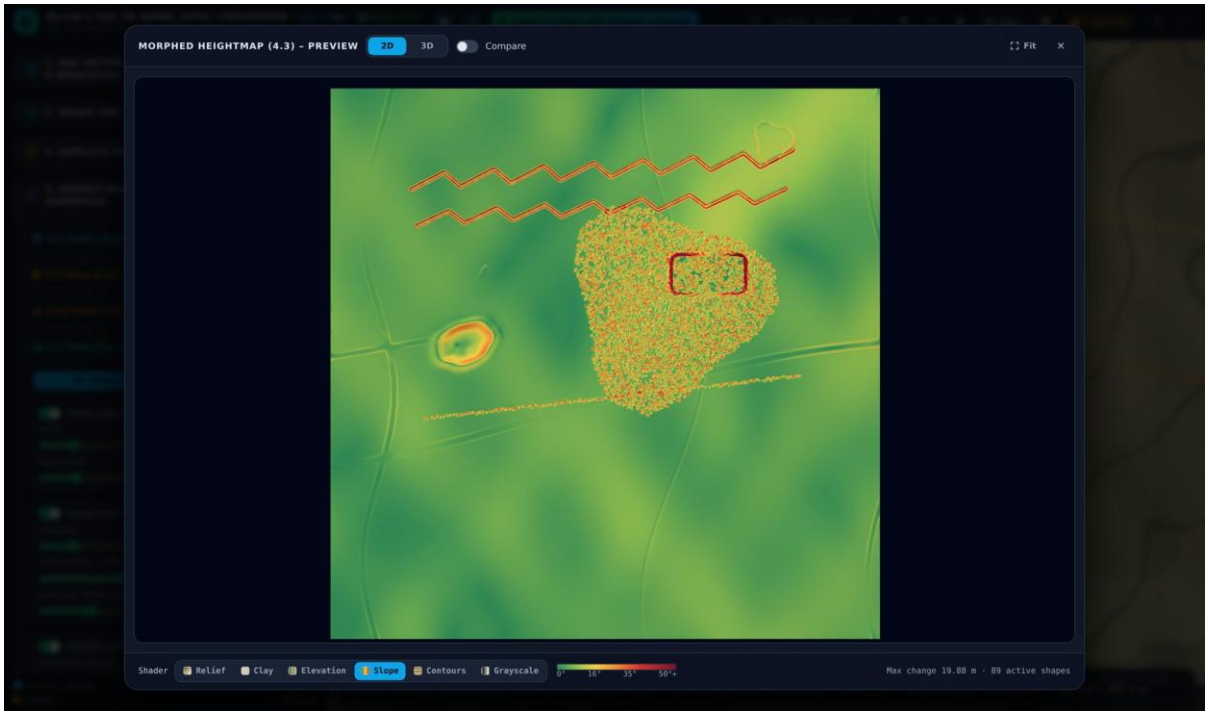


Figure 15 – *Slope shader: crater rims and trench walls in red*

Step 9 – Generate the surface masks

- Open 4.4. With the Reforger profile the splat masks are written per subtype and the Enfusion block check is on.
- Switch on “Slope & cliff auto-masking” to add rock on steep slopes. Preview compares the colour mask with the satellite image; blocks with too many surfaces are marked amber or red.

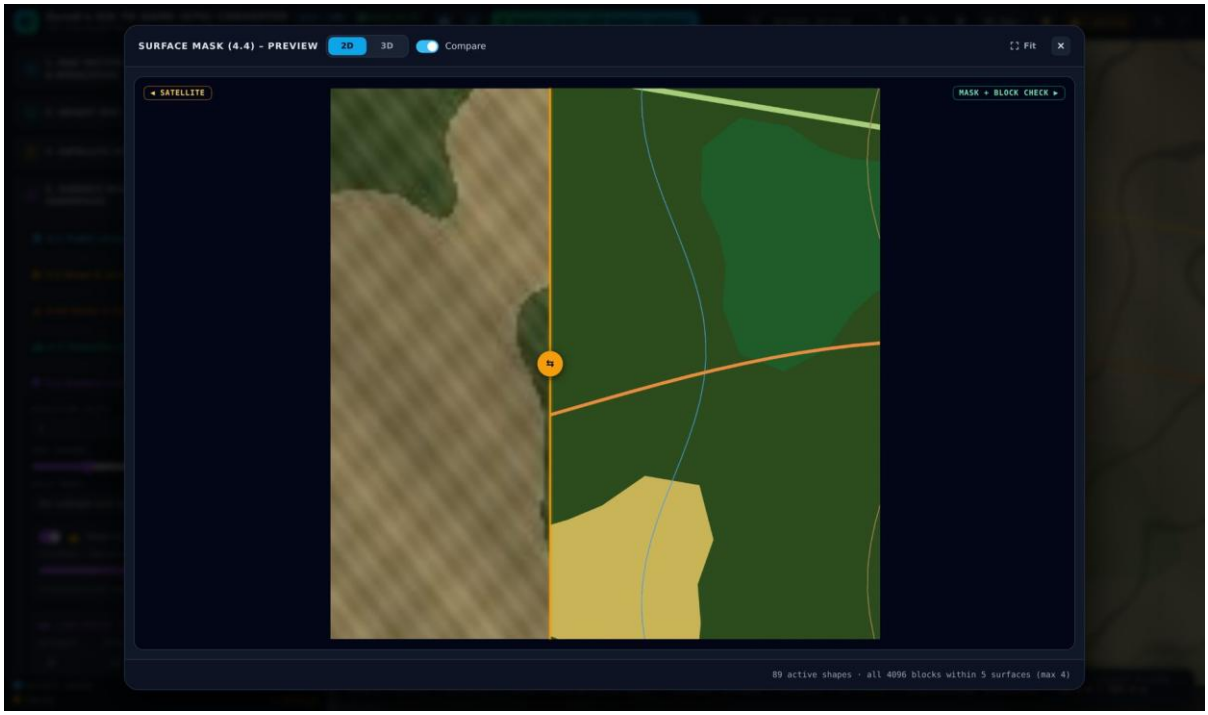


Figure 16 – *Colour-coded surface mask against the satellite image*

Step 10 – Export

- Click “Export sector with default settings” in the header for one ZIP with everything, or use the export button of each section for single parts.
- Large exports show an estimate first; a progress panel with Cancel follows.
- Use “Export params” in section 1 for the import values (height scale, grid, min / max height) – Arma Reforger needs them in the terrain import dialog.

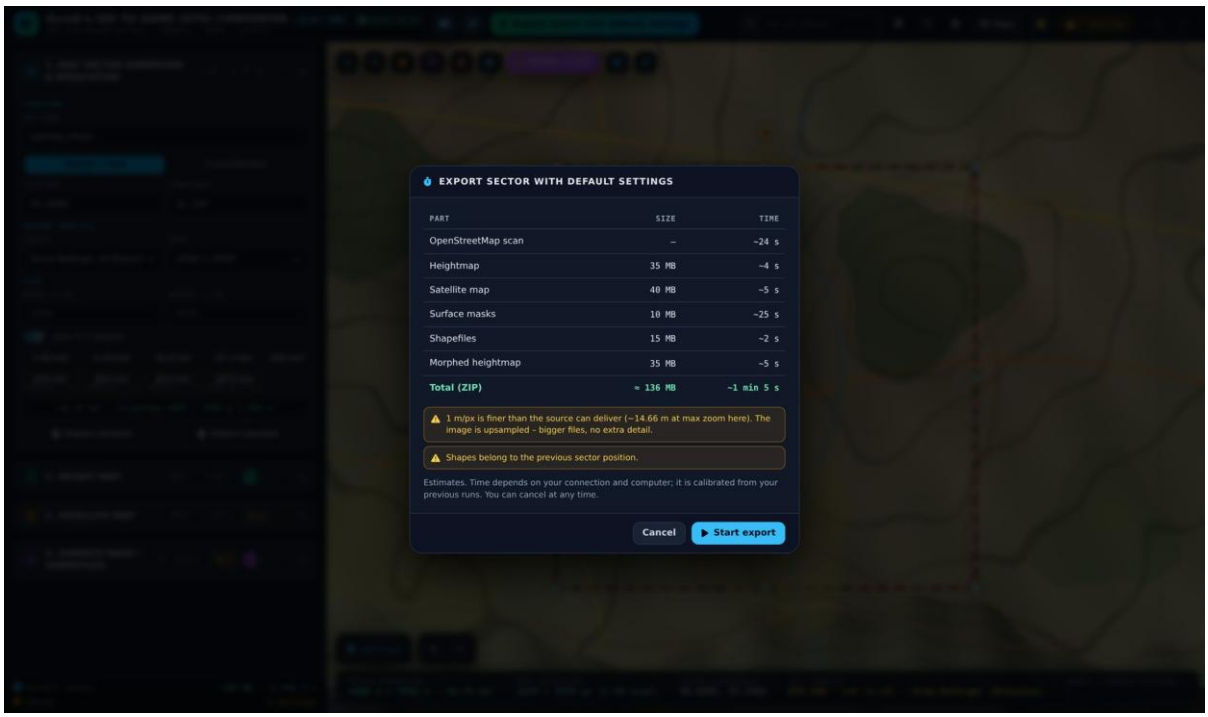


Figure 17 – *Export estimate before a large export*



Figure 18 – *Progress panel with elapsed / remaining time and Cancel*

What the default bundle contains

Folder	Size	Examples
1_Heightmap	7 files, 13.2 MB	, GTG_Lemnos_Front_1024x1024m_Heightmap.asc, GTG_Lemnos_Front_1024x1024m_Heightmap.prj, GTG_Lemnos_Front_1024x1024m_Heightmap_16bit.png, GTG_Lemnos_Front_1024x1024m_Heightmap_16bit.pgw, GTG_Lemnos_Front_1024x1024m_Heightmap_16bit.prj, …
2_Satellite	4 files, 588 KB	, GTG_Lemnos_Front_1024x1024m_Satellite.png, GTG_Lemnos_Front_1024x1024m_Satellite.pgw, GTG_Lemnos_Front_1024x1024m_Satellite.prj
3_SurfaceMask	30 files, 451 KB	, GTG_Lemnos_Front_1024x1024m_SurfaceMask.png, GTG_Lemnos_Front_1024x1024m_SurfaceMask.pgw, GTG_Lemnos_Front_1024x1024m_SurfaceMask.prj, GTG_Lemnos_Front_1024x1024m_SurfaceMask_legend.csv, GTG_Lemnos_Front_1024x1024m_Surface_01_lu_farmland.png, …
4_Shapefiles	33 files, 12 KB	, , GTG_Lemnos_Front_road_primary.shp, GTG_Lemnos_Front_road_primary.shx, GTG_Lemnos_Front_road_primary.dbf, GTG_Lemnos_Front_road_primary.cpg, …
5_Heightmap_Morphed	7 files, 13.2 MB	, GTG_Lemnos_Front_1024x1024m_Heightmap_Morphed.asc, GTG_Lemnos_Front_1024x1024m_Heightmap_Morphed.prj, GTG_Lemnos_Front_1024x1024m_Heightmap_Morphed_16bit.png, GTG_Lemnos_Front_1024x1024m_Heightmap_Morphed_16bit.pgw, GTG_Lemnos_Front_1024x1024m_Heightmap_Morphed_16bit.prj, …
(root)	2 files, 9 KB	GTG_Lemnos_Front_1024x1024m_params.json, ATTRIBUTION.txt

Importing the result into Arma Reforger

- Create a terrain entity at 0, 0, 0 with grid size 2048 × 2048 faces and cell size 1 m.
- Import the heightmap .asc (true metres) – or the 16-bit PNG with “Resample heights” and the min / max values from params.json.
- Import the satellite map and the grayscale surface masks (Surfaces_order.csv lists the stack order, 01 = bottom).
- Import the shapefiles from 4_Shapefiles/shp – they are in terrain metres, so they fit without offsets.

Part 3 – Feature reference

3.1 Section 1 – Map sector dimension & resolution

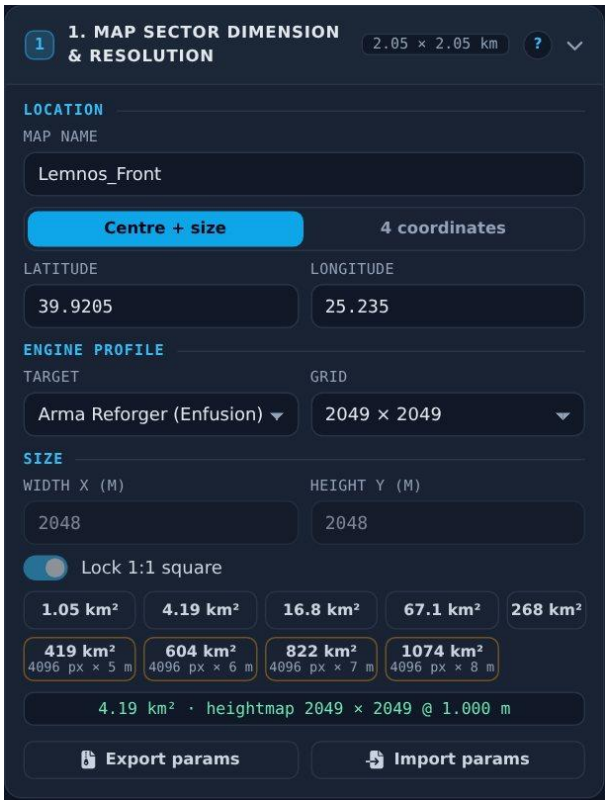


Figure 19 – *Section 1*

Location

- **Map name –** prefix of every exported file and ZIP.
- **Centre + size –** latitude / longitude of the sector centre (WGS84). Same as dragging the blue marker.
- **4 coordinates –** enter north, south, west, east or paste them (also four “lat, lon” corner lines from Google Maps or QGIS). The sector becomes the UTM rectangle that encloses them.

Engine profile

- **Target –** Custom, Unreal Engine 5, Unity, Arma 3, Arma Reforger. Sets grid, heightmap formats, satellite / mask resolution, vector format and coordinates.
- **Grid –** heightmap samples per side. Unreal / Unity / Reforger use vertex counts (2^n+1); Arma 3 uses cell counts (2^n). The sector size is (grid − 1) × cell size.

Size and presets

- **Width / height –** sector size in metres (fixed by the grid when a profile is active).
- **Lock 1:1 –** keeps the sector square.
- **Presets –** 1.05 km² up to 1,074 km². The large presets use a 4,096 px heightmap at 5–8 m cell size.
- **Export / import params –** saves or loads the whole project as .json, including all engine import values.

Direct manipulation on the map

- Blue centre marker: move the sector.
- Round corner handles: resize freely to any rectangle; hold Ctrl (or Alt) to keep it square.
- Green edge handles: change only width or height.
- Orange handle above the north edge: rotate (Ctrl = 5° steps).

Rotation and map grid

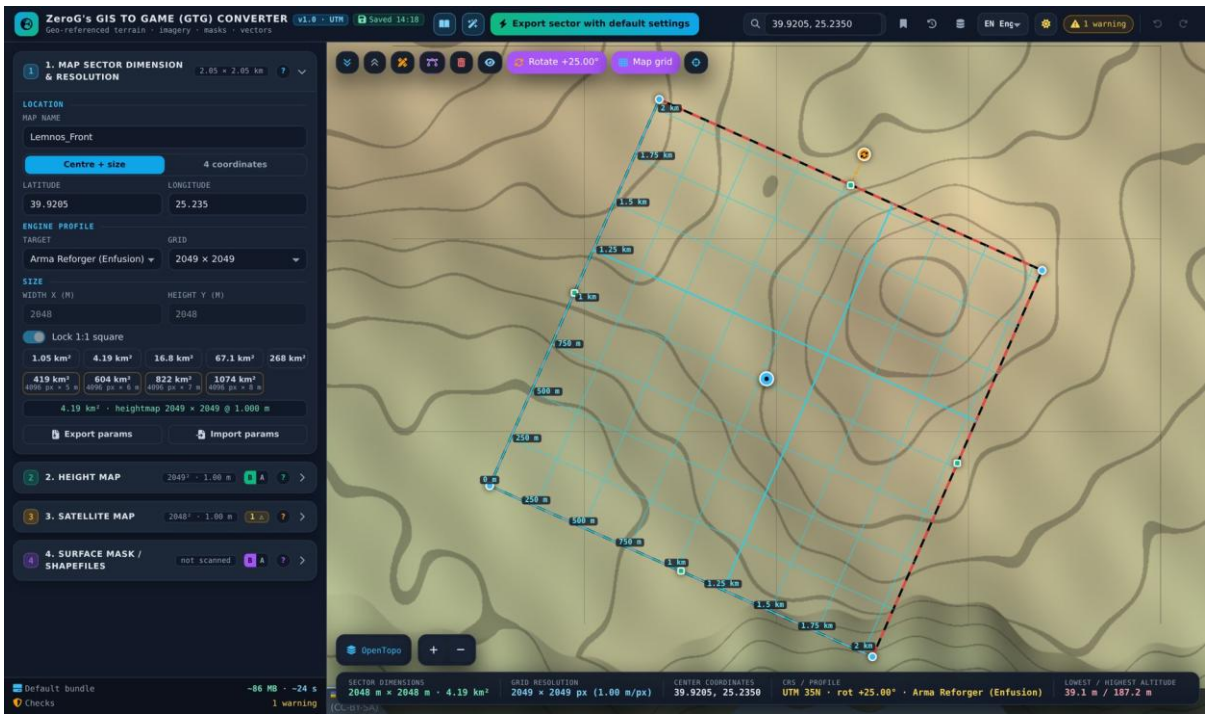


Figure 20 – *Sector rotated by 25°: the frame, the handles and all exports follow the rotation*

- **Rotate sector –** angle in degrees clockwise against UTM grid north; slider; ±0.1° / ±1° steps.
- **Align to map grid –** compensates the grid convergence so the edges run true north–south / east–west.
- **Align to UTM grid –** back to 0°.
- **Effect on exports –** heightmap, imagery and masks are resampled into the rotated grid; shapefiles in terrain metres follow the rotated axes; GeoTIFF and world files carry the rotation; .asc cannot store it (a message explains this).

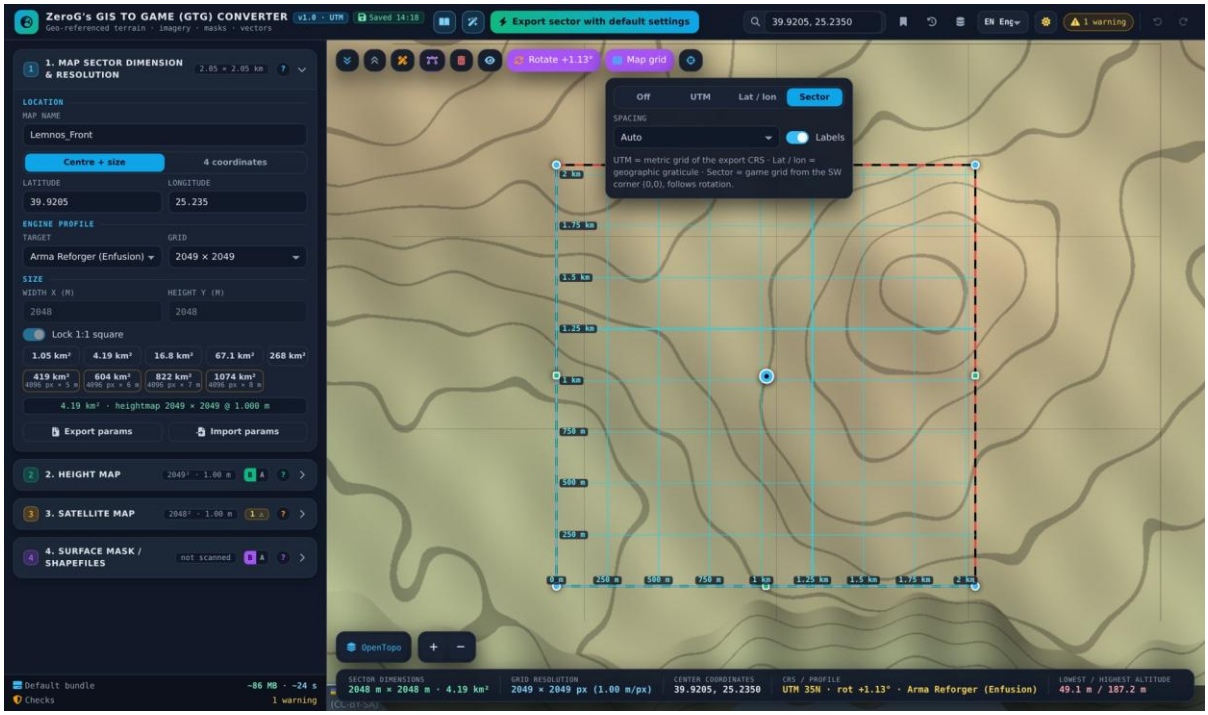


Figure 21 – *Map grid panel: Off, UTM, Lat / lon, Sector; spacing and labels*

- **UTM –** metric grid of the export CRS; every 10th line bold.
- **Lat / lon –** geographic graticule.
- **Sector –** game grid from the south-west corner (0,0); rotates with the sector.
- **Spacing –** Auto adapts to zoom or sector size; fixed 10 m – 10 km.

3.2 Section 2 – Height map

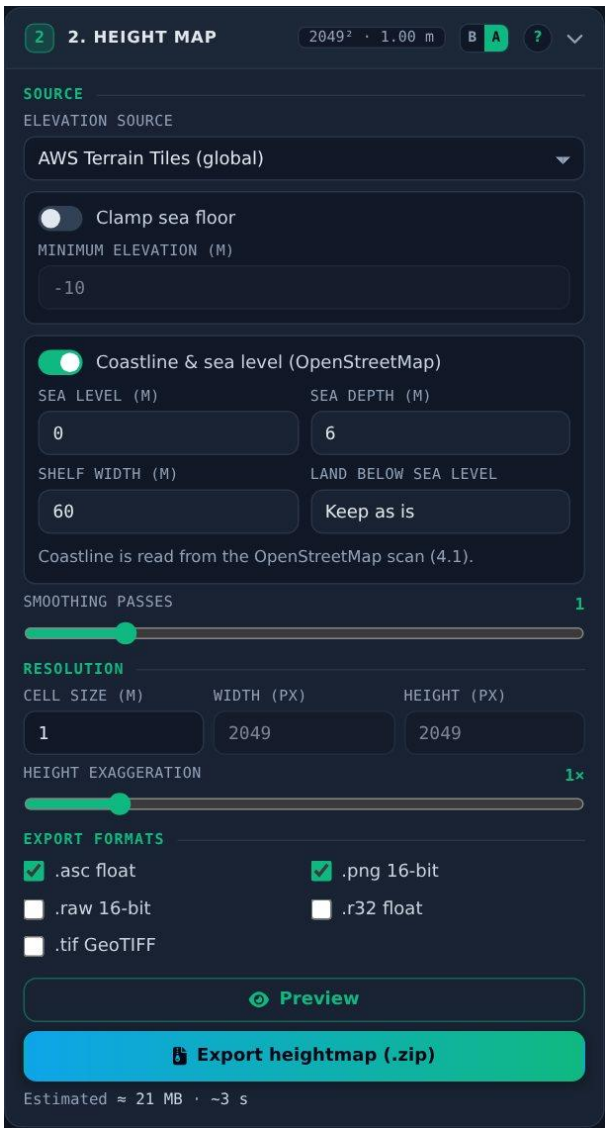


Figure 22 – *Section 2 in Advanced mode*

Elevation source

- **AWS Terrain Tiles –** free, global; tile grid ≈ 2–5 m, but survey data behind it is often 10–30 m outside lidar areas.
- **Custom WMS (elevation) –** official DEM servers; encodings GeoTIFF float32, Terrarium, Mapbox Terrain-RGB or grayscale with min / max. Gaps are filled from AWS.
- **Local DEM –** GeoTIFF or Esri ASCII in WGS84 or UTM. If it lies elsewhere, the sector moves onto it automatically.

Sea and coast

- **Clamp sea floor –** limits depth to save 16-bit range for land.
- **Coastline & sea level –** fills the sea side of OpenStreetMap coastlines; sea level, depth, shelf width; option to raise land below sea level.

Resolution and formats

- **Cell size –** 0.25–8 m. Width / height in px follow from sector ÷ cell size; typing pixels changes the cell size.
- **Smoothing passes –** light blur against terrace steps (1 is a good default).

- **Height exaggeration –** 0.25–5×, scaled from the lowest point; applies to previews and exports.
- **Formats –** .asc (true metres), .png 16-bit, .raw 16-bit (Unity / Unreal), .r32 float, .tif GeoTIFF (QGIS).

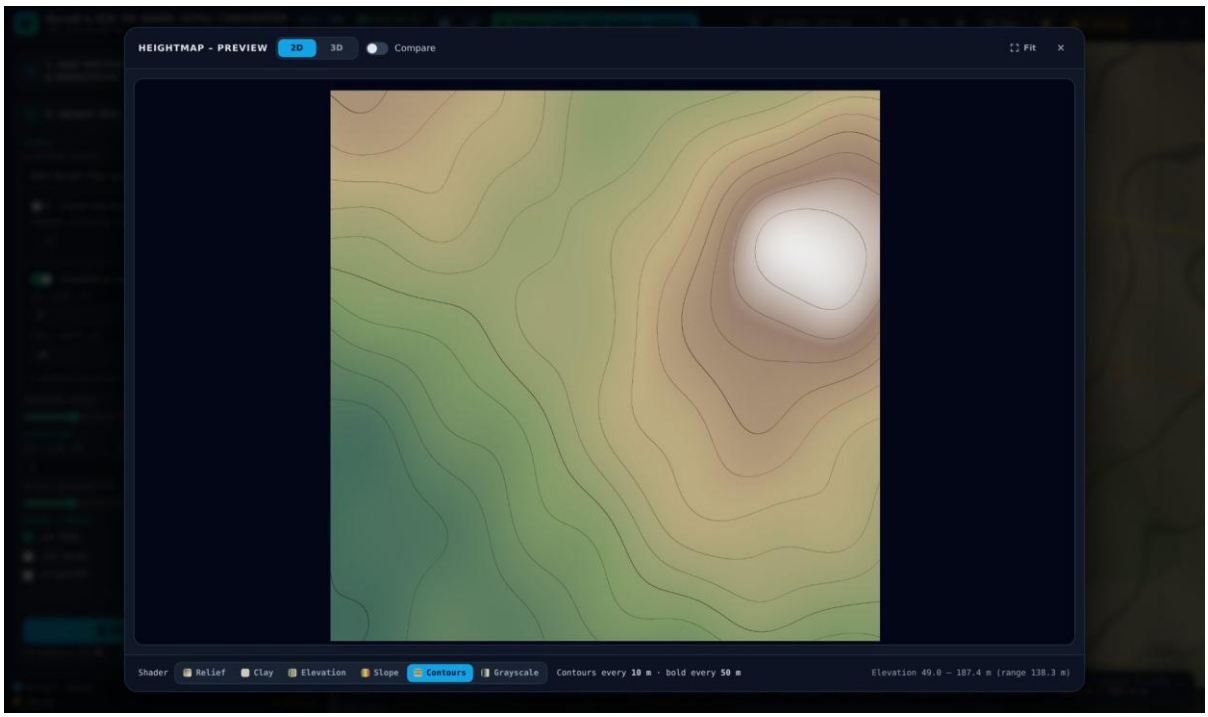


Figure 23 – *Contours shader – every 5th line bold, interval shown in the legend*

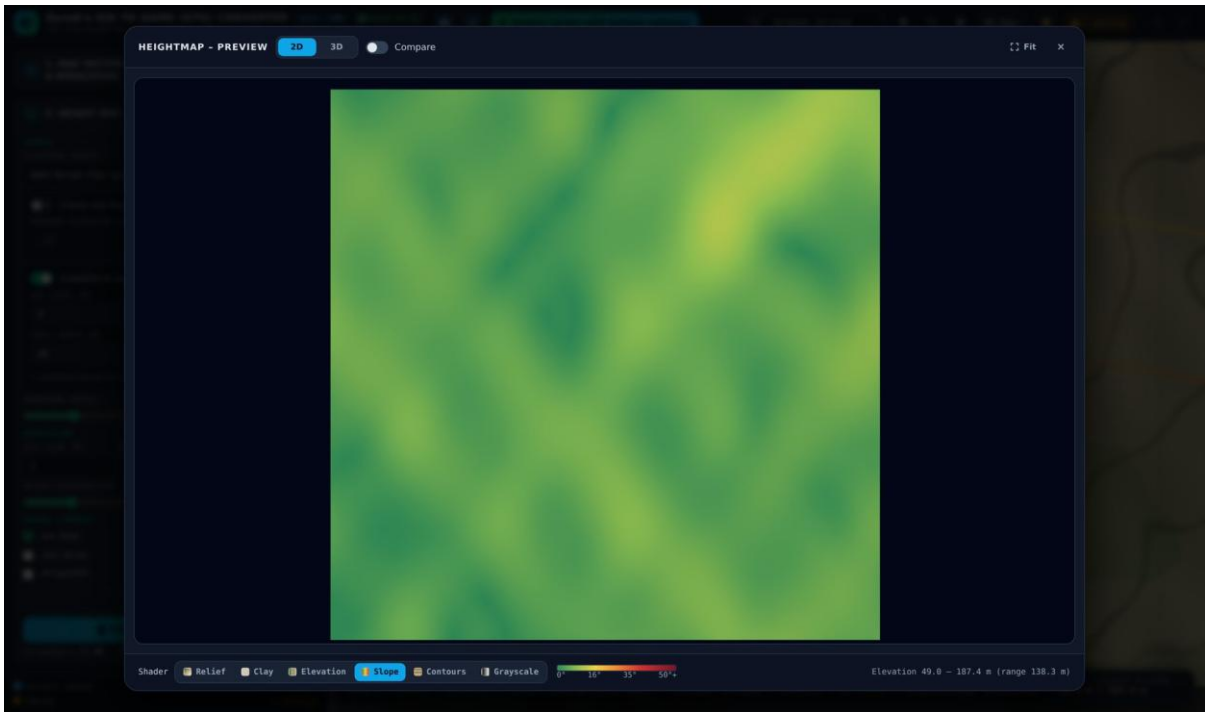


Figure 24 – *Slope shader – green flat, yellow ~16°, red > 35°*

3.3 Section 3 – Satellite map



Figure 25 – *Section 3*

- **Imagery source –** Sentinel-2 cloudless (10 m colour; 2016 = CC BY 4.0, 2018+ non-commercial), OpenTopoMap (rendered map), custom XYZ, custom WMS.
- **Resolution –** metres per pixel; a warning appears if you go finer than the source.
- **Width / height (px) –** linked to the sector size; typing pixels changes the resolution.
- **Format –** PNG (lossless) or JPG (about 5× smaller).

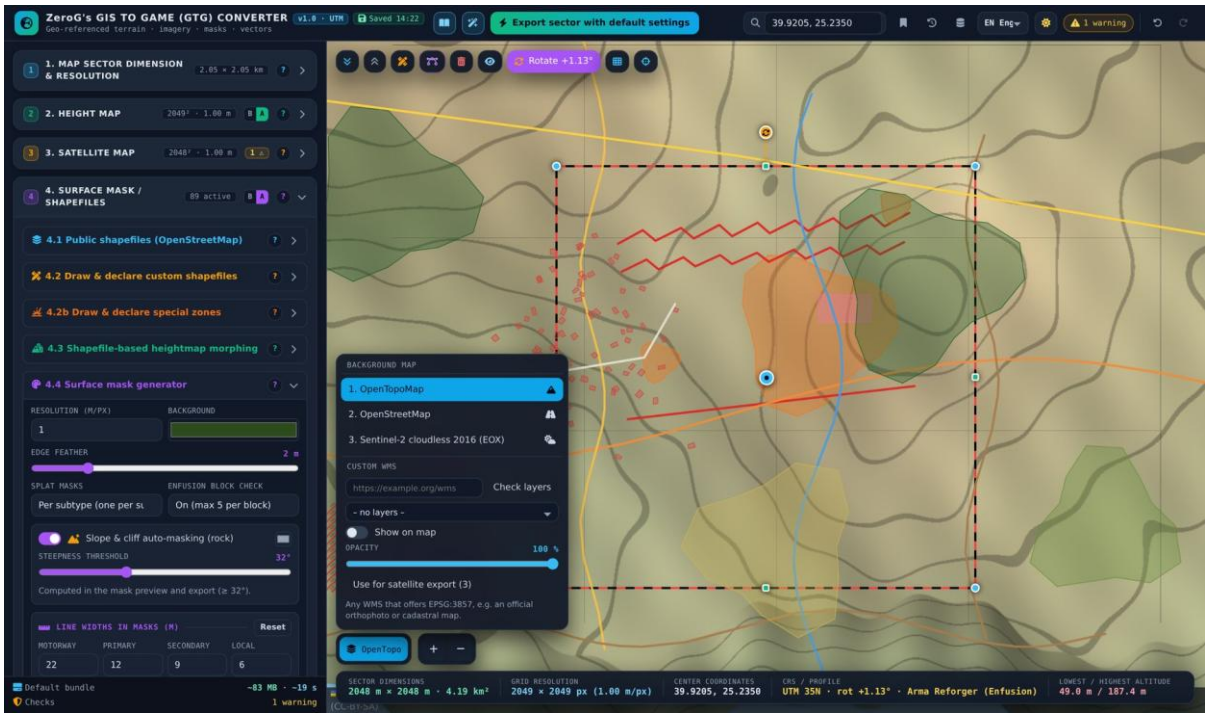


Figure 26 – *Background map menu with custom WMS overlay (opacity, "Use for satellite export")*

3.4 Section 4.1 – Public shapefiles (OpenStreetMap)

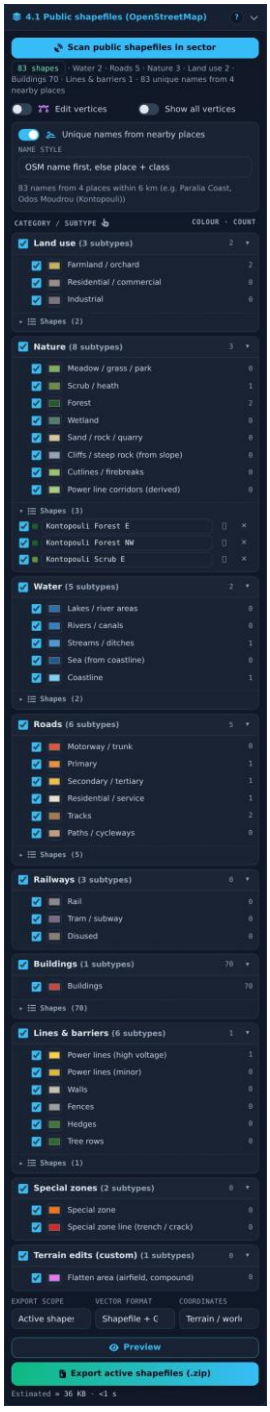


Figure 27 – *Section 4.1 with category tree and shape list*

Scan and names

- **Scan –** loads 33 subtypes in 8 categories; Shift+click forces a fresh download. A moved sector rescans automatically.
- **Unique names –** every shape gets a name from the nearest place, its class and direction (e.g. “Kontopouli Forest NE 02”). Written to NAME, OSM_NAME, PLACE, PLACE_DIR, PLACE_DIST.

Category tree and shape lists

- Tick categories / subtypes to show and export them; the colour sets map and mask colour.

- “Shapes (n)” lists every shape: include checkbox, name field, ⌖ zoom, ✕ delete. With 4.3 “Selected only” a target button picks shapes for morphing.
- Clicking a shape on the map excludes it (red diagonal bars); Ctrl+click selects it for 4.3.

Editing vertices

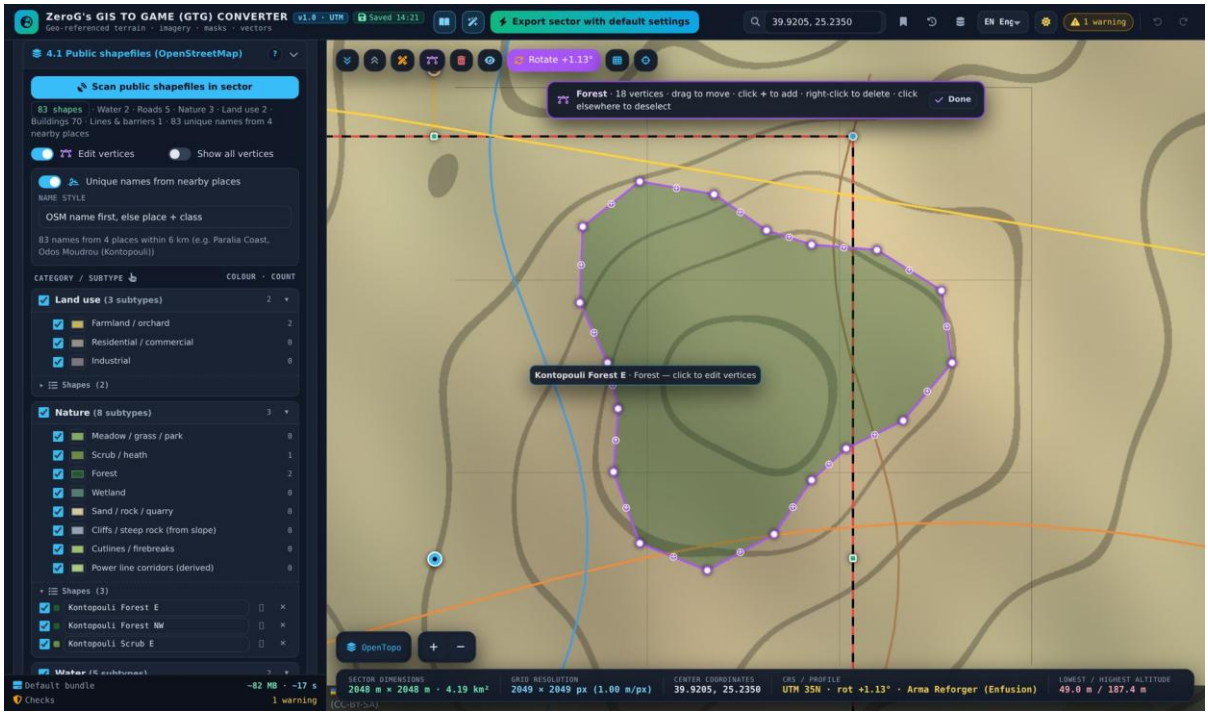


Figure 28 – *Edit vertices: drag a vertex, click + to add one, select and press Del to remove*

Export

- **Export scope –** active shapes only, or all shapes (deleted shapes never).
- **Vector format –** Shapefile + GeoJSON, Shapefile, GeoJSON.
- **Coordinates –** Terrain / world (metres from the south-west corner; use for engines) or UTM (for GIS). Shapes are cut to the terrain and kept 1 cm inside.

3.5 Section 4.2 – Custom shapefiles

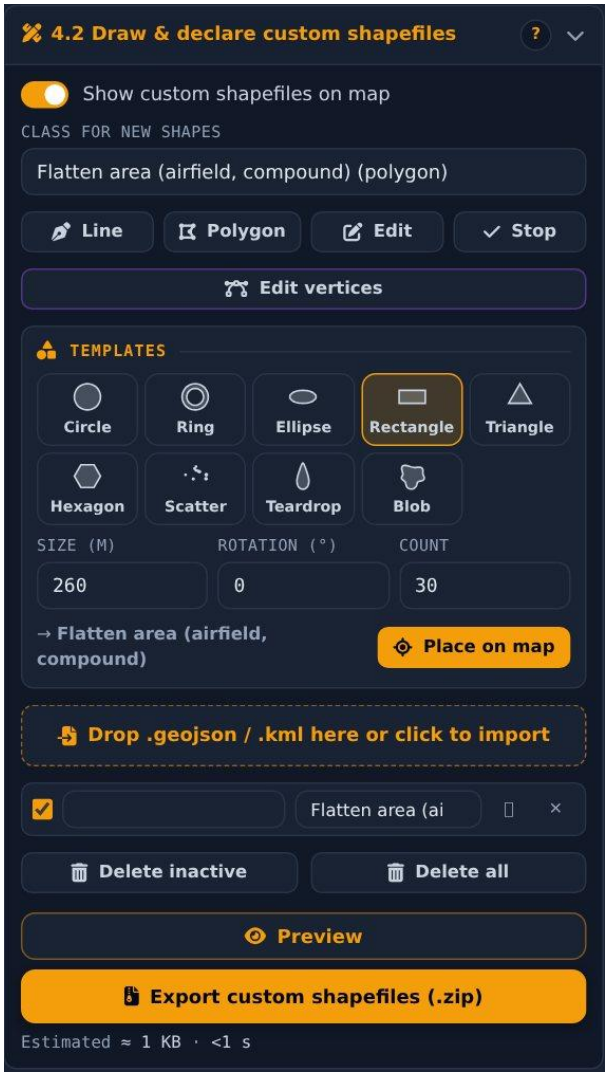


Figure 29 – *Section 4.2 with templates and shape list*

- **Class for new shapes –** all OSM classes plus “Flatten area” (airfields, compounds).
- **Draw tools –** Line, Polygon, Edit, Stop – or “Draw shapefiles” on the map toolbar (also Rectangle).
- **Templates –** circle, ring, ellipse, rectangle, triangle, hexagon, scatter, teardrop, blob; size, rotation and count; Shift+click places copies.
- **Import –** drop GeoJSON (WGS84, UTM, terrain metres) or KML; a “class” property is used when it matches.
- **Shape list –** include, rename, change class, zoom, delete; “Delete inactive” / “Delete all”.



Figure 30 – *Draw panel on the map toolbar*

3.6 Section 4.2b – Special zones

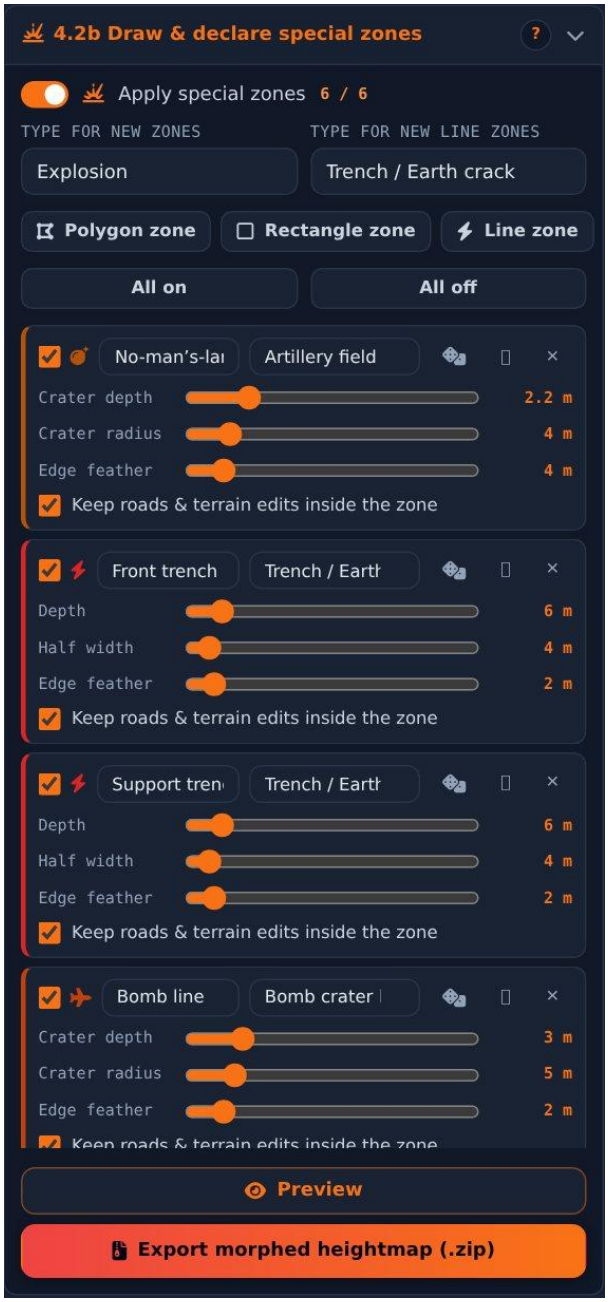


Figure 31 – *Section 4.2b: zone cards with individual settings*

- **Apply special zones –** master switch with active / total counter.
- **Type for new zones / line zones –** type given to the next drawn polygon or line.
- **Polygon zone, Rectangle zone, Line zone –** start drawing directly.
- **Zone card –** on/off, name, type, dice (new random layout), zoom, delete; sliders Intensity, Radius, Edge feather; “Keep roads & terrain edits inside the zone”.

Zone type	Shape	Effect
Explosion	Polygon	Single irregular crater with rim and one-sided ejecta.
Artillery field	Polygon	Many overlapping irregular craters – WW1 style.
Meteor impact	Polygon	Large crater with central peak and raised rim.
Sinkhole	Polygon	Steep-walled collapse.

Zone type	Shape	Effect
Volcano	Polygon	Cone with crater; wipes roads (keep off).
Erosion	Polygon	Gullies, stronger on steep slopes.
Firing position / revetment	Polygon	Dug-in pad with earth wall.
Dam / reservoir	Polygon	Raises a dam where the edge lies below the water level.
Open-pit mine / quarry	Polygon	Stepped benches down to a flat floor.
Mound / spoil heap	Polygon	Rounded dome with surface roughness.
Subsidence field	Polygon	Undulating sinking.
Dune field	Polygon	Asymmetric dunes; dice changes wind direction.
Rockfall / boulder field	Polygon	Scattered boulders.
Terraced fields	Polygon	Steps following the contours.
Trench / earth crack	Line / polygon	Narrow cut with noisy depth.
Bomb crater line	Line	Craters along the line.
Earthworks / berm	Line	Raised wall.
Anti-tank ditch	Line	Ditch with spoil berm on the right of the drawing direction.
Riverbed / dry wadi	Line	Meandering channel.

3.7 Section 4.3 – Heightmap morphing

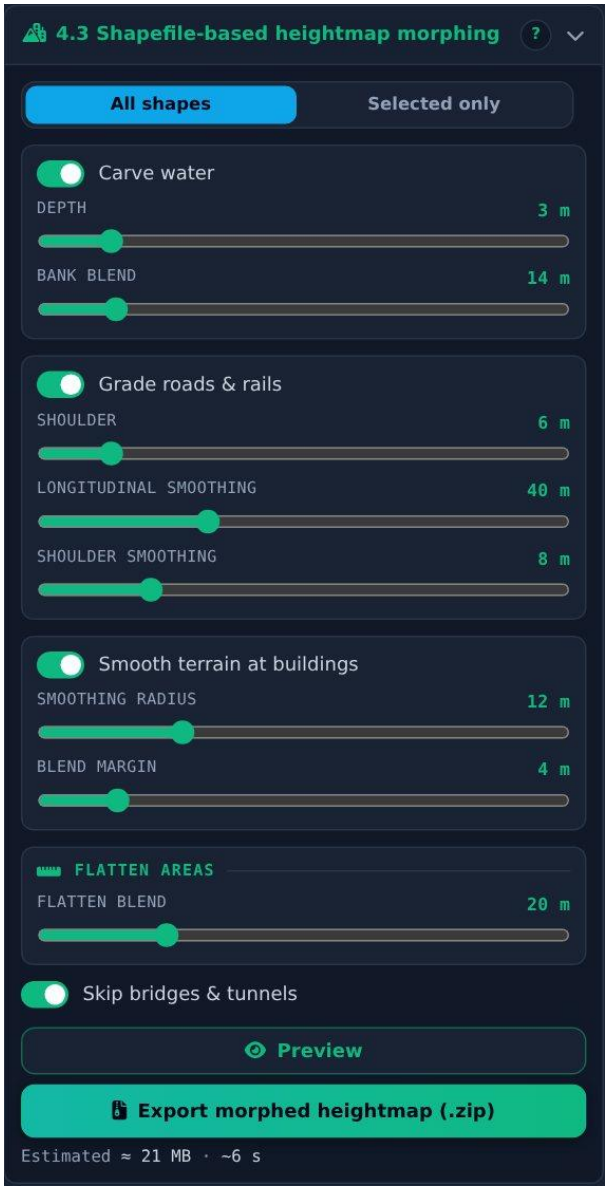


Figure 32 – *Section 4.3*

- **All shapes / Selected only –** apply shaping to everything or only to selected shapes.
- **Carve water –** flattens water to bank level, lowers it by Depth, blends over Bank blend.
- **Grade roads & rails –** flat cross-sections and smooth gradients; Shoulder, Longitudinal smoothing, Shoulder smoothing.
- **Smooth terrain at buildings –** smoothing radius and blend margin; slopes stay, bumps disappear.
- **Flatten areas –** each “Flatten area” polygon is levelled to its mean height.
- **Skip bridges & tunnels –** no dammed rivers, no cut hills.

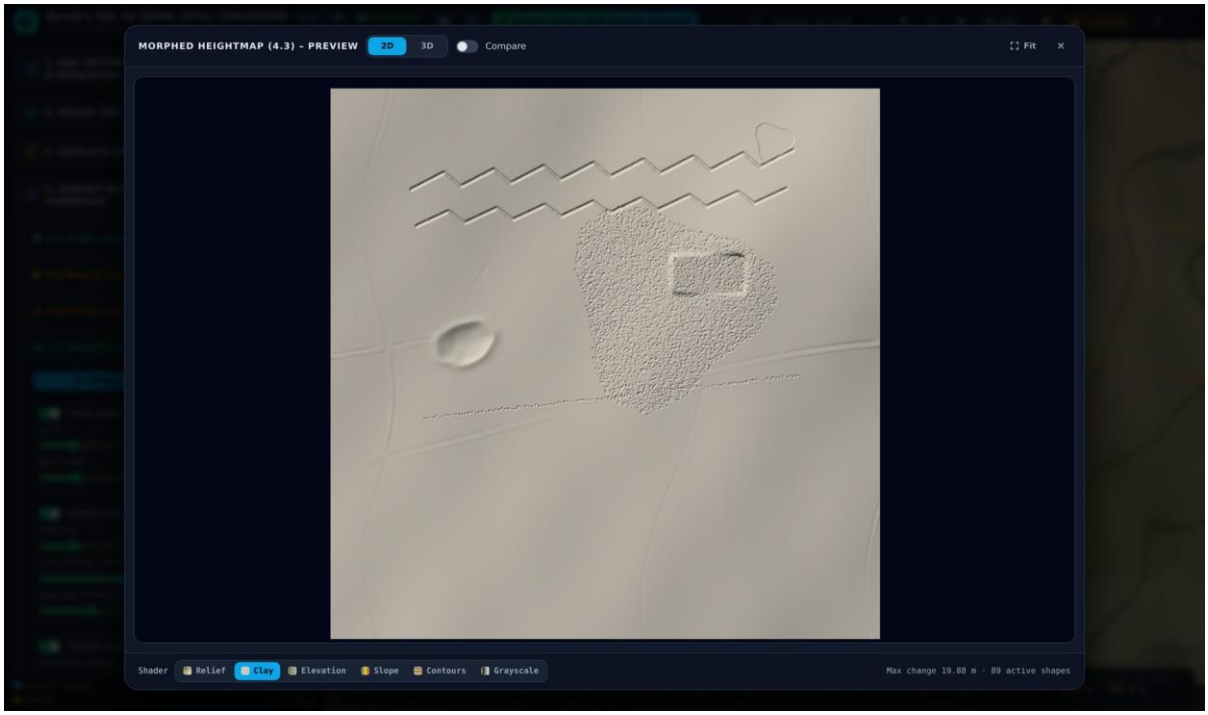


Figure 33 – *Clay shader – neutral model to judge the morphed shape*

3.8 Section 4.4 – Surface masks

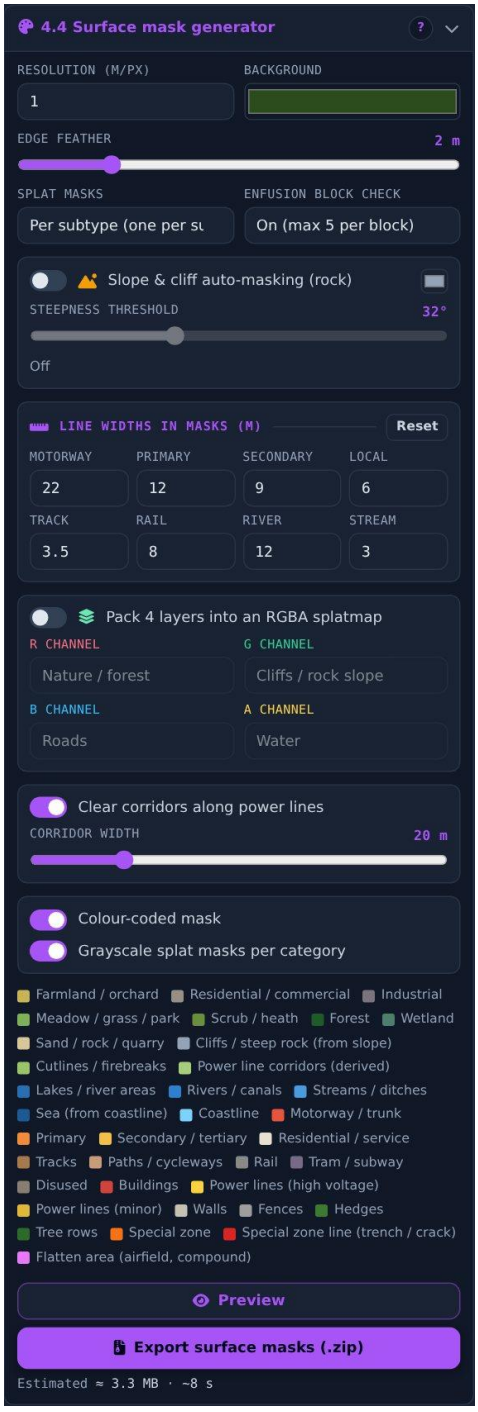


Figure 34 – *Section 4.4*

- **Resolution & background –** metres per mask pixel and colour of empty pixels.
- **Edge feather –** soft edges for grayscale splat masks (colour mask stays hard).
- **Splat masks –** per category or per subtype (one PNG per surface for Reforger), numbered in stack order.
- **Enfusion block check –** marks 32 × 32-face blocks with more than 5 surfaces; exports BlockCheck.png / .csv.
- **Slope & cliff auto-masking –** rock surface on slopes above the threshold (30–35° typical).
- **Line widths –** widths of roads, rails and rivers in masks and morphing.
- **RGBA splatmap –** four layers in one 8-bit PNG for Unity / Unreal.

- **Clearings –** corridors along power lines.
- **Outputs –** colour-coded mask with legend CSV and / or grayscale splat masks.

3.9 Preview window

- Mouse wheel zooms, drag pans, double-click fits; the orange handle moves the compare split.
- “Compare” switches the before / after view on or off.
- 3D tab: orbit with the mouse, extra height factor for viewing only.

Shader	Shows
Texture (3D)	Preview image draped over the terrain.
Relief	Coloured hillshade.
Clay	Neutral grey model – best for judging shape.
Elevation	Colour bands from low to high.
Slope	Green (flat) to red (> 35°).
Contours	Lines at an automatic interval, every 5th bold.
Grayscale	Heightmap values as the engine reads them.
Wireframe / Normals (3D)	Mesh density / surface direction.

3.10 Project management

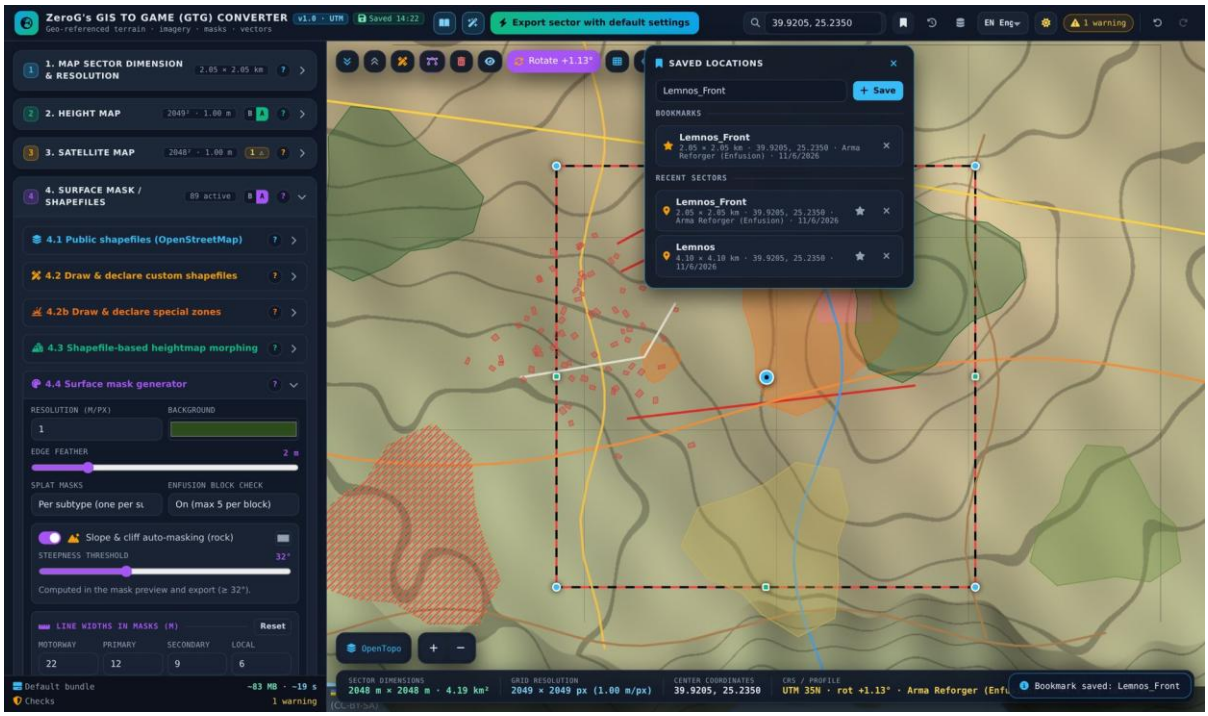


Figure 35 – *Locations: bookmarks and recent sectors*

- **Locations –** bookmark the current sector; recent sectors are recorded after each scan and export.
- **History –** last 20 versions: restore, compare with now, download as .json.

- **Data cache –** tiles and OpenStreetMap scans are kept between sessions; size limit and clearing.
- **Auto-save –** settings and shapes are saved while you work.

3.11 Language and theme



Figure 36 – *German interface*

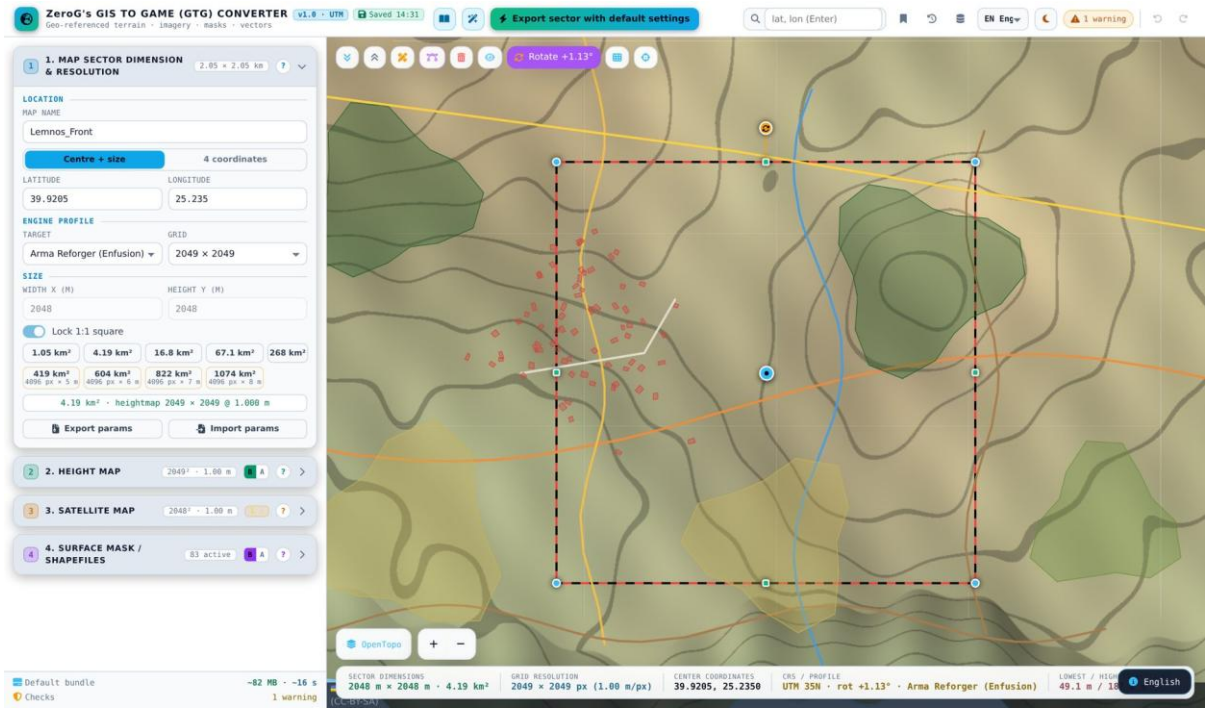


Figure 37 – *Bright theme*

Part 4 – Reference

4.1 Keyboard and mouse

Key / action	Function
Ctrl+Z / Ctrl+Y	Undo / redo
E	Edit vertices on / off
Del	Remove the selected vertex
Esc	Close dialog, cancel template, finish editing or delete mode
Click shape	Exclude / include in export
Ctrl+click shape	Select for 4.3 “Selected only”
Ctrl + drag corner	Keep the sector square
Ctrl + drag rotation handle	Rotate in 5° steps
Shift+click Scan	Force a fresh OpenStreetMap download
Shift+click map (template)	Place several copies
← / →	Wizard steps

4.2 Engine import notes

Engine	Key values
Arma Reforger	Terrain at 0,0,0; grid − 1 faces; .asc in true metres or 16-bit PNG with “Resample heights” (min / max in params). Height scale 0.03125 m = 2048 m range – params tells you if you need more. One grayscale PNG per surface; max 5 surfaces per block.
Arma 3	.asc heightmap, satellite and colour-coded mask with legend, UTM shapefiles.
Unreal Engine 5	16-bit PNG or RAW; X/Y scale = cell size × 100; Z scale from params.
Unity	RAW 16-bit, Windows byte order; terrain size and height from params.

4.3 Limits and formats

- Cell size 0.25–8 m; heightmap and images at most 8,192 px per side; sector at most 65,536 m.
- UTM is defined between 80° S and 84° N.
- OpenStreetMap scans above ~64 km² may time out at the Overpass servers.
- Many WMS servers block browser access (CORS) – the tool shows a message.

4.4 Troubleshooting

Problem	Solution
“Out of bounds” on shapefile import	Set 4.1 → Coordinates to “Terrain / world”.

Problem	Solution
WMS does not load	Server blocks browser requests (CORS) or lacks EPSG:3857 – try another server or a local file.
Special zone has no effect	“Apply special zones” on, zone ticked, section 4 in Advanced, preview the morphed heightmap (4.3).
Export is slow	Larger cell size, smaller sector; the estimate under every export button shows size and time.
Drawing does not start	The drawing library loads from unpkg.com – check internet access.
Rotated sector looks unrotated in GIS	Use the GeoTIFF; .asc cannot store rotation.

4.5 Credits and licences

- **OpenStreetMap –** © OpenStreetMap contributors, ODbL 1.0. Derived published data must stay under ODbL.
- **Mapzen Terrain Tiles on AWS –** USGS 3DEP, GMTED2010, SRTM; NOAA ETOPO1; EU-DEM (Copernicus) and regional sources.
- **Sentinel-2 cloudless –** s2maps.eu by EOX IT Services GmbH; 2016 CC BY 4.0, 2018+ CC BY-NC-SA 4.0.
- **OpenTopoMap –** map data © OpenStreetMap contributors, SRTM; style © OpenTopoMap (CC-BY-SA).
- **Software –** Leaflet (BSD-2), Leaflet-Geoman Free, JSZip, geotiff.js, three.js (MIT); Font Awesome Free; Inter and JetBrains Mono (OFL).
- Every export ZIP contains ATTRIBUTION.txt with the required credit text.